EMRE KÜÇÜKGÖÇER

Applied AI, Machine Learning & Data Science Engineer | Computer Vision, GenAI & MLOps

Phone: 0543 721 22 82 • **Email:** emrekucukgocer42@gmail.com • **LinkedIn:** linkedin.com/in/emrekucukgocer

• **GitHub:** github.com/emrekgcc

To: Hiring Team & Engineering Leadership **Date:** October 31, 2026

Dear Hiring Manager,

As a senior-year Computer Engineering student at Sivas Cumhuriyet University (graduating June 2027) and a certified graduate of the **Samsung Innovation Campus Artificial Intelligence Program**, I am writing to apply for an engineering role within your organization. I do not merely train isolated machine learning models in notebooks; I architect, deploy, and govern end-to-end production systems. My background bridges real-time Edge AI, autonomous Multi-Agent LLM orchestration, classical credit risk and decision science, and resilient MLOps infrastructure.

Through my industry internships, award-winning engineering projects, and rigorous applied AI training, I bring immediate, production-tested value across four core engineering pillars:

- **Generative AI, Enterprise RAG & Autonomous Multi-Agent Systems:** During my internship at **HT Bilişim**, I spearheaded a production-grade Enterprise RAG architecture utilizing semantic chunking, hybrid search, reranking, and vector databases (Pinecone/Milvus) to guarantee hallucination-free retrieval on proprietary data. I also architected **Council AI**, a Clean Architecture educational ecosystem powered by a sequential **LangGraph** pipeline (Socrates, Fact Checker, and Quest Master agents) and an asynchronous **FastAPI** backend, reinforced by LLMOps observability, Guardrails, Red Teaming, and prompt-injection defenses.
- **Computer Vision, Edge AI & Medical AI R&D:** In **DentAI (1st Place, TAÜ Hackathon 2026)**, I engineered a privacy-preserving MedTech platform by deploying a custom **YOLOv8-Nano** model directly on-device via **TensorFlow Lite**. By combining hard negative mining with an **OpenCV CLAHE** optical preprocessing pipeline and a custom 3D-printed macro apparatus, I achieved **88.8% field accuracy** with zero cloud latency. Currently, as a Medical AI R&D Intern at **Ratione Divide (Radia AI)**, I develop explainable clinical decision-support systems and medical-imaging workflows.
- **Data Science, Financial Risk Modeling & Big Data Analytics:** Beyond deep learning, I possess a strong foundation in classical machine learning (**Scikit-Learn, XGBoost, LightGBM**), Advanced SQL, and statistical experimentation (A/B testing, RFM segmentation). I am well-versed in regulated banking and risk analytics—including **Credit Scoring, PD/LGD/EAD modeling, IFRS 9 principles**, imbalanced data handling, probability calibration, and Explainable AI (**SHAP/LIME**)—complemented by big data and streaming architectures (**Apache Spark, Kafka, Databricks, Snowflake, BigQuery**).
- **High-Throughput Backend, Multi-Tenant Systems & MLOps:** At **Uçak Yazılım**, I built a distributed OSINT data pipeline with Python asynchronous workers, complex ETL workflows, and a zero-latency FastAPI microservice layer. In my **TÜBİTAK-funded Health Management System**, I engineered a multi-tenant Flutter and Firebase architecture with RBAC isolation and offline-resilient asynchronous data queueing. Furthermore, I operationalize AI pipelines using **Docker, Kubernetes, CI/CD, MLflow (Model Registry)**, and continuous data/concept drift monitoring.

Whether your team is scaling enterprise LLM agents, deploying low-latency computer vision models at the edge, or validating high-impact risk and decision-science models, I have the hands-on engineering discipline to deliver clean, scalable, and governance-compliant code from day one.

I have attached my resume for your review and would welcome the opportunity to discuss how my end-to-end engineering background aligns with your technical roadmap. Thank you for your time and consideration.

Sincerely,

Emre Küçükgöçer

Computer Engineering Student | Applied AI, ML & Data Science Engineer Candidate